

Chapter 5 – Regression Analysis

Chapter 5 – Regression Analysis

Linear • Multiple • Polynomial • Logistic • Ridge • Lasso • Elastic Net

Chapter Learning Objectives

Understand regression as a supervised-learning method for predicting continuous variables.

Distinguish correlation, regression and causation.

Construct simple linear regression models and estimate coefficients.

Use matrix notation for linear and multiple regression.

Evaluate regression models using MSE, RMSE, Relative MSE, coefficient of variation and coefficient of determination.

Model nonlinear relationships using transformations and polynomial regression.

Understand logistic regression for binary classification.

Explain ridge, Lasso and Elastic Net regularization.

5.1 Introduction to Regression

Regression is used to predict continuous or quantitative variables such as price and revenue.

Regression analysis seeks answers to: What is the relationship between variables? How strong is it? Is it linear or nonlinear?

It also asks whether attributes are relevant and what contribution each attribute makes.

Applications listed in the book include sales prediction, bond values, insurance premiums, crop yield and real-estate prices.

Regression: Typical Problem Structure

Independent / predictor variables: explanatory attributes used for prediction.

Dependent / response variable: quantitative variable to be predicted.

Regression estimates a mathematical relationship between predictors and the response



Regression and Correlation

Correlation describes the strength and direction of association between variables.

A scatter plot is a useful visual tool for exploring the relationship.

Positive correlation: variables tend to increase together.

Negative correlation: one variable tends to decrease as the other increases.

No correlation: no clear systematic relationship is visible.

Regression vs Correlation vs Causation

Regression / Correlation

Regression focuses on prediction of one variable from another.

Correlation measures strength/direction of association.

Neither by itself proves cause-and-effect.

Causation

Causation asks whether one variable causes a change in another.

Correlation and regression relationships are not automatically causal relationships.

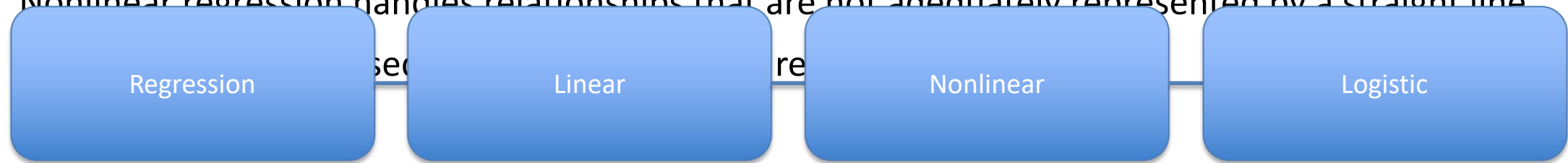
Domain knowledge and appropriate study design are required for causal conclusions.

Types of Regression Methods

The book's classification includes linear regression, nonlinear regression and logistic regression.

Linear regression fits a linear relationship.

Nonlinear regression handles relationships that are not adequately represented by a straight line.



5.3 Introduction to Linear Regression

The simplest linear regression model fits a line through scattered data points.

Model: $y = a_0 + a_1x + \varepsilon$.

a_0 is the intercept; a_1 is the slope; ε is the prediction error.

The coefficients are chosen so that the fitted line provides the best fit under the chosen criterion.

$$\hat{y} = a_0 + a_1x$$

$a_0 = \text{intercept}; a_1 = \text{slope}$

Assumptions of Linear Regression

The book introduces assumptions associated with linear regression before deriving the fitted line.

Observations of y are treated as random.

The relationship between predictor and response is represented through a linear model.

Residuals capture the difference between observed and fitted values.

For practical modeling, inspect residuals and the data relationship rather than assuming linearity automatically.

Residuals and Best-Fit Line

Residual for observation i : $e_i = y_i - \hat{y}_i$.

Residuals may be positive, negative or zero.

A regression line is a line of best fit when the sum of squared residuals is minimized.

This is the Ordinary Least Squares (OLS) principle used in the chapter.

$$\text{Minimize } \sum (y_i - \hat{y}_i)^2$$

OLS chooses coefficients that minimize the sum of squared residuals.

Simple Linear Regression Coefficients

For one predictor, the slope can be computed from sample means and cross-products.

The intercept is then obtained from the mean relationship.

The resulting line can be used for interpolation/extrapolation with appropriate caution.

$$a_1 = [\Sigma x_i y_i - n \bar{x} \bar{y}] / [\Sigma x_i^2 - n \bar{x}^2]$$

$$a_0 = \bar{y} - a_1 \bar{x}$$

Worked Example from the Textbook

Data used in the book: $x = 1, 2, 3, 4, 5$ and $y = 1.2, 1.8, 2.6, 3.2, 3.8$.

Computed summaries: $\Sigma x = 15$, $\Sigma y = 12.6$, $\Sigma x^2 = 55$, $\Sigma xy = 44.4$.

$\bar{x} = 3$ and $\bar{y} = 2.52$.

Computed coefficients: $a_1 \approx 0.66$ and $a_0 \approx 0.54$

$$\hat{y} = 0.54 + 0.66x$$

Fitted regression line: $\hat{y} = 0.54 + 0.66x$.

Textbook example: predicted 7th-week value ≈ 5.16 ; 12th-month value ≈ 8.46 .

Linear Regression – Interpretation of Coefficients

Intercept a_0 : predicted response when $x = 0$, if that value is meaningful within the application.

Slope a_1 : expected change in the predicted response for a one-unit increase in x .

Positive slope $\rightarrow$ increasing linear trend.

Negative slope $\rightarrow$ decreasing linear trend.

The numerical meaning depends on the units of x and y .

Linear Regression in Matrix Form

The model can be written compactly using matrices.

Y is the $n \times 1$ response vector.

X is the design matrix containing a column for the intercept and columns for predictors.

a is the vector of regression coefficients. **$Y = Xa + \varepsilon$**

ε is the $n \times 1$ error vector.

OLS coefficient estimate: $\hat{a} = (X^T X)^{-1} X^T Y$, when the inverse exists.

Multiple Linear Regression

Multiple regression uses more than one independent variable to predict a response.

General form: $y = a_0 + a_1x_1 + a_2x_2 + \dots + a_px_p + \varepsilon$.

Each coefficient represents the contribution of a predictor while the other predictors are included in the model.

$$\hat{y} = a_0 + \sum a_j x_j$$

Matrix formulation provides a convenient way to estimate all coefficients together.

$$j = 1, \dots, p$$

Multiple Regression – Matrix Estimation

Construct X by including a column of 1s for the intercept and one column for each predictor.

Construct Y from observed response values.

Compute $X^T X$ and $X^T Y$.

Solve the normal-equation system to obtain coefficient vector a .

$$\hat{a} = (X^T X)^{-1} X^T Y$$

The chapter illustrates the calculation using a sample dataset.

Check dimensions: X is $n \times (p+1)$, a is $(p+1) \times 1$, Y is $n \times 1$.

5.4 Validation of Regression Methods

Regression models should be evaluated using metrics that compare predicted and actual values.

The chapter discusses MSE, RMSE, Relative MSE, coefficient of variation, standard error estimate and coefficient of determination.

No single metric tells the whole story; interpretation should consider scale, variability and the application.

Mean Squared Error (MSE)

MSE is the average of squared prediction errors.

Squaring prevents positive and negative residuals from canceling.

MSE penalizes larger errors more strongly than smaller errors.

Lower MSE indicates better fit under this metric.

$$\text{MSE} = (1/n) \sum (y_i - \hat{y}_i)^2$$

Units are squared units of the response variable.

Root Mean Squared Error (RMSE)

RMSE is the square root of MSE.

It returns the error to the same units as the response variable.

The book uses RMSE to summarize the typical scale of prediction error.

Example shown in the text: if $MSE = 62.5$, $RMSE = \sqrt{62.5} \approx 7.91$.

Example: $\sqrt{62.5} \approx 7.91$

Relative MSE

Relative MSE compares prediction error against the variability of a trivial/population reference.

The book notes that a value of zero indicates a perfect model under this measure.

A value greater than 1 indicates that the created model is not good according to the chapter's interpretation.

Use relative measures when comparing performance across contexts with different scales.

Coefficient of Variation (CV)

Coefficient of variation compares standard deviation with the mean.

It is useful for comparing variability across datasets measured on different scales.

The chapter discusses CV as one of the regression-evaluation measures.

Interpret the value together with the units and mean of the response.

$$CV = \sigma / \mu$$

$$\text{Percentage CV} = (\sigma / \mu) \times 100$$

Coefficient of Determination R^2

R^2 summarizes how much of the variation in the response is accounted for by the regression model.

It is commonly interpreted as the square of the correlation between predicted and response values in the simple linear setting discussed in the chapter.

Higher R^2 indicates stronger explanatory fit, but it should not be treated as proof of causation.

R^2 should be interpreted alongside residual error and the model context.

$$R^2 = 1 - \text{SSE}/\text{SST}$$

$$\text{SSE} = \sum (y_i - \hat{y}_i)^2; \text{SST} = \sum (y_i - \bar{y})^2$$

Standard Error Estimate

The chapter illustrates standard error estimation using the squared differences between observed and predicted values.

It quantifies the residual spread around the fitted regression relationship.

Use it to understand the typical variability left unexplained by the regression model.

Always distinguish residual standard error from standard errors of individual coefficients.

5.6 Nonlinear / Polynomial Regression

Some relationships are not adequately represented by a straight line.

The chapter discusses transformations for nonlinear relationships and polynomial regression.

Transformations such as logarithms can convert certain nonlinear relationships into a form suitable for linear regression.

Polynomial regression introduces powers of x to model curvature.

Transformation-Based Regression

Exponential relationships can be transformed using logarithms.

Power relationships can also be transformed: $y = ax^b$ becomes $\log y = b \log x + \log a$.

After fitting the transformed linear relationship, inverse transformations are applied to obtain the desired result.

$$\text{Power model: } y = ax^b \rightarrow \log y = b \log x + \log a$$

Transformation should be chosen according to the form of the relationship.

Transform $\rightarrow$ fit linear model $\rightarrow$ inverse-transform

Polynomial Regression

Polynomial regression models nonlinear relationships using powers of the predictor.

Quadratic: $y = a_0 + a_1x + a_2x^2$.

Cubic: $y = a_0 + a_1x + a_2x^2 + a_3x^3$.

The book notes that high-degree polynomials can create very flexible curves and may lead to overfitting; it mentions practical use of degrees up to about four.

$$\hat{y} = a_0 + a_1x + a_2x^2 + \dots + a_mx^m$$

Higher degree $\rightarrow$ more flexibility $\rightarrow$ greater risk of overfitting.

Polynomial Regression – Matrix Form

For a degree-m polynomial, the design matrix contains columns $[1, x, x^2, \dots, x^m]$.

The coefficients are estimated using the same least-squares idea as linear regression.

The model is linear in the coefficients even though it is nonlinear in x .

The chapter provides a second-degree worked example using a matrix system.

$$X = [1, x, x^2, \dots, x^m]$$

$$\hat{a} = (X^T X)^{-1} X^T Y$$

5.7 Logistic Regression

Linear regression predicts numerical/continuous responses and is not appropriate for categorical responses.

Logistic regression is suitable for binary classification.

Examples from the chapter: spam/not spam, admitted/not admitted, pass/fail.

The model predicts the probability of a categorical outcome.

Why Not Use Linear Regression for Probability?

Linear regression can produce values below 0 or above 1.

Probabilities must lie between 0 and 1.

Logistic regression therefore maps a linear predictor into the interval [0,1].

The chapter uses the sigmoid/logit formulation to make this mapping.

$$p(x) = 1 / [1 + \exp(-(a_0 + a_1x))]$$

Sigmoid output is between 0 and 1.

Odds and Log-Odds (Logit)

Odds compare the probability of an event with the probability of the event not occurring.

Odds = $p / (1-p)$.

Log-odds (logit) = $\log[p/(1-p)]$.

Logistic regression models the log-odds as a linear function of predictor variables.

$$\text{logit}(p) = \log[p/(1-p)] = a_0 + a_1x$$

$$p = \exp(a_0 + a_1x) / [1 + \exp(a_0 + a_1x)]$$

Logistic Classification Using a Threshold

After computing $p(y=1 | x)$, a threshold can convert probability into a class label.

The chapter gives a threshold of 0.5 as an example.

If predicted probability > 0.5 , classify as class 1; otherwise classify as class 0.

Threshold selection changes the balance between false positives and false negatives.

$$\hat{y} = 1 \text{ if } p(x) > 0.5, \text{ otherwise } \hat{y} = 0$$

Threshold can be tuned for the application.

Worked Logistic Example from the Text

The chapter gives a binary student pass/fail-style example based on entrance marks.

Regression coefficients are used to compute the linear predictor for $x = 60$.

The resulting sigmoid probability is approximately 0.44 in the example.

With threshold 0.5, $0.44 < 0.5$, so the candidate is classified as not selected.

The example illustrates the complete chain: coefficients $\rightarrow$ linear score $\rightarrow$ sigmoid probability $\rightarrow$ thresholded class.

Learning Logistic Regression Parameters

The chapter describes parameter estimation using Maximum Likelihood Estimation (MLE).

The objective is to learn coefficient values for which the observed data has maximum likelihood under the logistic model.

The log-likelihood is commonly optimized because products of probabilities become sums after taking logs.

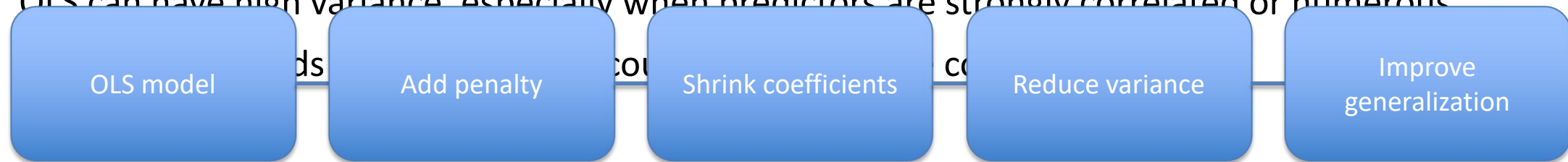
The idea extends beyond binary classification to multinomial logistic regression.

5.8 Ridge, Lasso and Elastic Net

Regularization is introduced to control model complexity and reduce test error.

The chapter motivates regularization using the bias–variance trade-off.

OLS can have high variance, especially when predictors are strongly correlated or numerous.



Bias–Variance Motivation

Underfitting

Bias: error associated with a model that does not fit the data well.

Model too simple

Variance: variation in model predictions across different training datasets.

High bias

High variance with low bias can lead to overfitting.

Low variance

High bias with low variance can lead to underfitting.

Poor training fit

Regularization deliberately adds some bias to reduce variance and improve test performance.

Overfitting

Bias: error associated with a model that does not fit the data well.

Model too flexible

Low training error

High variance

Poor unseen-data performance

5.8.1 Ridge Regularization

Ridge regression is a regularized linear regression method.

It is useful when there are many attributes and strong correlations among them.

Instead of estimating coefficients only by OLS, ridge uses a biased estimator with lower variance.

The penalty discourages large regression coefficients.

$$\text{Minimize: } \sum (y_i - \hat{y}_i)^2 + \lambda \sum a_j^2$$

L2 penalty; λ controls the strength of regularization.

Ridge: Matrix Form and λ

The chapter gives a matrix formulation for ridge regression.

The regularized coefficient estimate modifies $X^T X$ by adding a penalty term.

λ controls the amount of shrinkage.

Large $\lambda \rightarrow$ stronger shrinkage and more bias; small $\lambda \rightarrow$ closer to OLS.

Choosing λ is therefore an important model-selection step.

Choose λ using validation / re-sampling.

5.8.2 Lasso Regression

Lasso is another regularization method.

It adds an L1 penalty to the regression objective.

The L1 penalty can shrink some coefficients all the way to zero.

Therefore, Lasso can perform coefficient shrinkage and implicit feature selection.

It is useful when many predictors may be irrelevant.

$$\text{Minimize: } \sum (y_i - \hat{y}_i)^2 + \lambda \sum |a_j|$$

L1 penalty $\rightarrow$ sparse coefficient vector.

5.8.3 Elastic Net

Elastic Net combines L1 and L2 regularization ideas.

It can shrink parameters associated with correlated variables and may set irrelevant variables to zero.

The chapter highlights its usefulness when multicollinearity exists among independent variables.

Elastic Net therefore combines sparsity from L1 with stabilizing shrinkage from L2.

$$\text{Loss} + \lambda_1 \sum |a_j| + \lambda_2 \sum a_j^2$$

Combination of L1 and L2 penalties.

Ridge vs Lasso vs Elastic Net

Ridge (L2)

Shrinks coefficients toward zero

Usually retains all predictors

Useful with correlated predictors

Controls variance

Lasso / Elastic Net

Lasso: L1 can set coefficients to zero

Elastic Net: combines L1 + L2

Useful for feature selection and multicollinearity

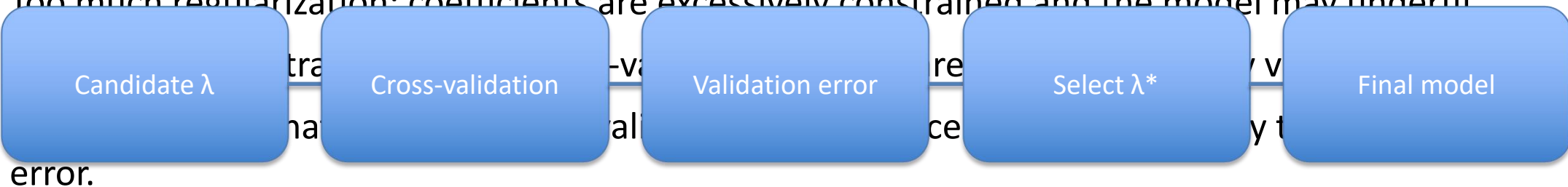
Both require penalty tuning

Choosing the Regularization Strength

The penalty parameter controls the bias–variance trade-off.

Too little regularization: model remains flexible and may overfit.

Too much regularization: coefficients are excessively constrained and the model may underfit



End-to-End Regression Workflow

1. Define the response and predictor variables.
2. Explore scatter plots/correlations and understand the domain.
3. Fit a suitable regression model.



6. Add nonlinear terms or regularization when justified.
7. Test the final model on unseen data and interpret coefficients carefully.

Important Formula Sheet

Simple regression: $\hat{y} = a_0 + a_1x$.

Slope: $a_1 = [\sum xy - n \bar{x}\bar{y}] / [\sum x^2 - n \bar{x}^2]$.

Intercept: $a_0 = \bar{y} - a_1\bar{x}$.

MSE = $(1/n)\sum(y-\hat{y})^2$; RMSE = $\sqrt{\text{MSE}}$.

$R^2 = 1 - \text{SSE}/\text{SST}$.

Logistic: $p = 1/[1+e^{-z}]$, $z = a_0 + a_1x$ (or $a^T x$).

Ridge: squared-error + $\lambda \sum a^2$; Lasso: squared-error + $\lambda \sum |a|$; Elastic Net: both penalties.

Exam-oriented Questions

1. Explain regression analysis and distinguish regression, correlation and causation.
2. Derive the simple linear regression coefficients and solve a numerical problem.
3. Explain OLS and compute MSE/RMSE/ R^2 for given actual and predicted values.
4. Explain matrix-form linear/multiple regression.
5. Explain transformation-based and polynomial regression with a worked example.
6. Explain logistic regression, sigmoid, odds and log-odds.
7. Explain ridge, Lasso and Elastic Net and compare L1 vs L2 regularization.

Chapter 5 – Key Takeaways

Regression is a supervised-learning approach for prediction and relationship modeling.

Linear regression provides the foundational OLS model.

Multiple regression extends the model to several predictors.

Validation metrics quantify prediction error and goodness of fit.

Polynomial/transform-based methods model nonlinear relationships.

Logistic regression maps a linear predictor to a probability for binary classification.

Regularization controls complexity: Ridge shrinks, Lasso can select, Elastic Net combines both.